

Homework 3: STLC — Solutions

Problem 1

$$\tau = \mathbf{T} \rightarrow \mathbf{T}$$

Typing

$$\frac{\mathcal{D}_{\text{scrut}} \quad \mathcal{D}_{\text{left}} \quad \mathcal{D}_{\text{right}}}{\bullet \vdash M : \mathbf{T} \rightarrow \mathbf{T}} \text{ (CASE)}$$

Here $\mathcal{D}_{\text{scrut}}$ denotes the following derivation tree:

$$\frac{\frac{\frac{\mathcal{D}_{\text{id}} \quad \mathcal{D}_{\text{id}}}{\bullet \vdash \langle \lambda x : \mathbf{T}. x, \lambda x : \mathbf{T}. x \rangle : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T})} \text{ (PAIR)}}{\bullet \vdash \text{right} \langle \lambda x : \mathbf{T}. x, \lambda x : \mathbf{T}. x \rangle : (\mathbf{T} \rightarrow \mathbf{T}) + ((\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (RIGHT)}$$

Use $\mathcal{D}_{\text{id}}$ to denote the following derivation tree:

$$\frac{\frac{x : \mathbf{T} \in \{x : \mathbf{T}\}}{x : \mathbf{T} \vdash x : \mathbf{T}} \text{ (VAR)}}{\bullet \vdash \lambda x : \mathbf{T}. x : \mathbf{T} \rightarrow \mathbf{T}} \text{ (ABS)}$$

Use $\mathcal{D}_{\text{left}}$ to denote the following derivation tree:

$$\frac{\frac{f : \mathbf{T} \rightarrow \mathbf{T} \in \{f : \mathbf{T} \rightarrow \mathbf{T}\}}{f : \mathbf{T} \rightarrow \mathbf{T} \vdash f : \mathbf{T} \rightarrow \mathbf{T}} \text{ (VAR)}}{\bullet \vdash \lambda f : \mathbf{T} \rightarrow \mathbf{T}. f : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (ABS)}$$

Let

$$\Gamma = p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}), x : \mathbf{T}$$

Use $\mathcal{D}_{\text{right}}$ to denote the following derivation tree:

$$\frac{\frac{\frac{p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}) \in \Gamma}{\Gamma \vdash p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T})} \text{ (VAR)}}{\Gamma \vdash \text{proj1 } p : \mathbf{T} \rightarrow \mathbf{T}} \text{ (PROJ1)} \quad \mathcal{D}_{\text{inner}} \text{ (APP)}}{\Gamma \vdash (\text{proj1 } p)((\text{proj2 } p)x) : \mathbf{T}} \text{ (ABS)}$$

$$\frac{p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}) \vdash \lambda x : \mathbf{T}. (\text{proj1 } p)((\text{proj2 } p)x) : \mathbf{T} \rightarrow \mathbf{T}}{\bullet \vdash \lambda p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}). \lambda x : \mathbf{T}. (\text{proj1 } p)((\text{proj2 } p)x) : ((\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T})) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (ABS)}$$

Use $\mathcal{D}_{\text{inner}}$ to denote the following derivation tree:

$$\frac{\frac{\frac{p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}) \in \Gamma}{\Gamma \vdash p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T})} \text{ (VAR)}}{\Gamma \vdash \text{proj2 } p : \mathbf{T} \rightarrow \mathbf{T}} \text{ (PROJ2)} \quad \frac{x : \mathbf{T} \in \Gamma}{\Gamma \vdash x : \mathbf{T}} \text{ (VAR)}}{\Gamma \vdash (\text{proj2 } p)x : \mathbf{T}} \text{ (APP)}$$

Reduction

$$\begin{aligned} M &\longrightarrow (\lambda p : (\mathbf{T} \rightarrow \mathbf{T}) \times (\mathbf{T} \rightarrow \mathbf{T}). \lambda x : \mathbf{T}. (\mathbf{proj1}\ p)((\mathbf{proj2}\ p)x)) \\ &\quad \langle \lambda x : \mathbf{T}. x, \lambda x : \mathbf{T}. x \rangle \\ &\longrightarrow \lambda x : \mathbf{T}. (\mathbf{proj1}\ \langle \lambda x : \mathbf{T}. x, \lambda x : \mathbf{T}. x \rangle) \\ &\quad ((\mathbf{proj2}\ \langle \lambda x : \mathbf{T}. x, \lambda x : \mathbf{T}. x \rangle)x) \\ &\longrightarrow \lambda x : \mathbf{T}. (\lambda x : \mathbf{T}. x)((\mathbf{proj2}\ \langle \lambda x : \mathbf{T}. x, \lambda x : \mathbf{T}. x \rangle)x) \\ &\longrightarrow \lambda x : \mathbf{T}. (\lambda x : \mathbf{T}. x)((\lambda x : \mathbf{T}. x)x) \\ &\longrightarrow \lambda x : \mathbf{T}. (\lambda x : \mathbf{T}. x)x \\ &\longrightarrow \lambda x : \mathbf{T}. x \end{aligned}$$

Problem 2

(a)(i) Typing

$$\tau = T \rightarrow T$$

The typing derivation is:

$$\frac{\mathcal{D}_{\text{pair}} \quad \mathcal{D}_{\text{body}}}{\bullet \vdash M : T \rightarrow T} \text{ (LET-PAIR)}$$

Here $\mathcal{D}_{\text{pair}}$ denotes the following derivation tree:

$$\frac{\mathcal{D}_{\text{first}} \quad \mathcal{D}_{\text{second}}}{\bullet \vdash \langle (\lambda h : T \rightarrow T. h)(\lambda a : T. a), \lambda b : T. b \rangle : (T \rightarrow T) \times (T \rightarrow T)} \text{ (PAIR)}$$

Use $\mathcal{D}_{\text{first}}$ to denote the following derivation tree:

$$\frac{\frac{\frac{h : T \rightarrow T \in \{h : T \rightarrow T\}}{h : T \rightarrow T \vdash h : T \rightarrow T} \text{ (VAR)} \quad \frac{\frac{a : T \in \{a : T\}}{a : T \vdash a : T} \text{ (VAR)}}{\bullet \vdash \lambda h : T \rightarrow T. h : (T \rightarrow T) \rightarrow (T \rightarrow T)} \text{ (ABS)} \quad \frac{\frac{a : T \vdash a : T}{} \text{ (ABS)}}{\bullet \vdash \lambda a : T. a : T \rightarrow T} \text{ (ABS)} \quad \frac{}{\bullet \vdash (\lambda h : T \rightarrow T. h)(\lambda a : T. a) : T \rightarrow T} \text{ (APP)}$$

Use $\mathcal{D}_{\text{second}}$ to denote the following derivation tree:

$$\frac{\frac{b : T \in \{b : T\}}{b : T \vdash b : T} \text{ (VAR)}}{\bullet \vdash \lambda b : T. b : T \rightarrow T} \text{ (ABS)}$$

Let

$$\Gamma_1 = f : T \rightarrow T, g : T \rightarrow T$$

Use $\mathcal{D}_{\text{body}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{function}} \quad \frac{g : T \rightarrow T \in \Gamma_1}{\Gamma_1 \vdash g : T \rightarrow T} \text{ (VAR)}}{\Gamma_1 \vdash (\lambda k : T \rightarrow T. \lambda z : T. f(kz)) g : T \rightarrow T} \text{ (APP)}$$

Let

$$\Gamma_2 = \Gamma_1, k : T \rightarrow T, z : T$$

Use $\mathcal{D}_{\text{function}}$ to denote the following derivation tree:

$$\frac{\frac{\frac{f : T \rightarrow T \in \Gamma_2}{\Gamma_2 \vdash f : T \rightarrow T} \text{ (VAR)} \quad \frac{\frac{k : T \rightarrow T \in \Gamma_2}{\Gamma_2 \vdash k : T \rightarrow T} \text{ (VAR)} \quad \frac{z : T \in \Gamma_2}{\Gamma_2 \vdash z : T} \text{ (VAR)}}{\Gamma_2 \vdash kz : T} \text{ (APP)}}{\Gamma_2 \vdash f(kz) : T} \text{ (APP)} \quad \frac{}{\Gamma_1, k : T \rightarrow T \vdash \lambda z : T. f(kz) : T \rightarrow T} \text{ (ABS)} \quad \frac{}{\Gamma_1 \vdash \lambda k : T \rightarrow T. \lambda z : T. f(kz) : (T \rightarrow T) \rightarrow (T \rightarrow T)} \text{ (ABS)}$$

Problem 2 (continued)

(a)(ii) Reduction

The first pair component must reduce before let elimination; no rule reduces the let body.

$$\begin{aligned} M &\longrightarrow \text{let } \langle f, g \rangle = \langle \lambda a : \mathbf{T}. a, \lambda b : \mathbf{T}. b \rangle \\ &\quad \text{in } (\lambda k : \mathbf{T} \rightarrow \mathbf{T}. \lambda z : \mathbf{T}. f(k\ z))\ g \\ &\longrightarrow (\lambda k : \mathbf{T} \rightarrow \mathbf{T}. \lambda z : \mathbf{T}. (\lambda a : \mathbf{T}. a)(k\ z))\ (\lambda b : \mathbf{T}. b) \\ &\longrightarrow \lambda z : \mathbf{T}. (\lambda a : \mathbf{T}. a)((\lambda b : \mathbf{T}. b)\ z) \\ &\longrightarrow \lambda z : \mathbf{T}. (\lambda a : \mathbf{T}. a)\ z \\ &\longrightarrow \lambda z : \mathbf{T}. z \end{aligned}$$

Problem 2 (continued)

(b)(i) Typing

$$\tau = (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))$$

The typing derivation is:

$$\frac{\mathcal{D}_{\text{outer}} \quad \mathcal{D}_{\text{nested}}}{\bullet \vdash \text{let } \langle f, g \rangle = \langle \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h, \lambda x : \mathbf{T}. x \rangle \text{ in} \\ \text{let } \langle k, p \rangle = \langle f, \langle g, f \rangle \rangle \text{ in } \langle k(\text{proj1 } p), \text{proj2 } p \rangle : \\ (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (LET-PAIR)}$$

Here $\mathcal{D}_{\text{outer}}$ denotes the following derivation tree:

$$\frac{\mathcal{D}_{\text{fun}} \quad \mathcal{D}_{\text{id}}}{\bullet \vdash \langle \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h, \lambda x : \mathbf{T}. x \rangle : \\ ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})) \times (\mathbf{T} \rightarrow \mathbf{T})} \text{ (PAIR)}$$

Use $\mathcal{D}_{\text{fun}}$ to denote the following derivation tree:

$$\frac{\frac{h : \mathbf{T} \rightarrow \mathbf{T} \in \{h : \mathbf{T} \rightarrow \mathbf{T}\} \text{ (VAR)}}{h : \mathbf{T} \rightarrow \mathbf{T} \vdash h : \mathbf{T} \rightarrow \mathbf{T}} \text{ (VAR)}}{\bullet \vdash \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (ABS)}$$

Use $\mathcal{D}_{\text{id}}$ to denote the following derivation tree:

$$\frac{\frac{x : \mathbf{T} \in \{x : \mathbf{T}\} \text{ (VAR)}}{x : \mathbf{T} \vdash x : \mathbf{T}} \text{ (VAR)}}{\bullet \vdash \lambda x : \mathbf{T}. x : \mathbf{T} \rightarrow \mathbf{T}} \text{ (ABS)}$$

Let

$$\Gamma_1 = f : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}), g : \mathbf{T} \rightarrow \mathbf{T}$$

Use $\mathcal{D}_{\text{nested}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{inner}} \quad \mathcal{D}_{\text{body}}}{\Gamma_1 \vdash \text{let } \langle k, p \rangle = \langle f, \langle g, f \rangle \rangle \\ \text{in } \langle k(\text{proj1 } p), \text{proj2 } p \rangle : \\ (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (LET-PAIR)}$$

Use $\mathcal{D}_{\text{inner}}$ to denote the following derivation tree:

$$\frac{\frac{f : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}) \in \Gamma_1}{\Gamma_1 \vdash f : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (VAR)} \quad \mathcal{D}_{\text{pack}}}{\Gamma_1 \vdash \langle f, \langle g, f \rangle \rangle : \\ ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})) \\ \times ((\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})))} \text{ (PAIR)}$$

Use $\mathcal{D}_{\text{pack}}$ to denote the following derivation tree:

$$\frac{\frac{g : \mathbf{T} \rightarrow \mathbf{T} \in \Gamma_1}{\Gamma_1 \vdash g : \mathbf{T} \rightarrow \mathbf{T}} \text{ (VAR)} \quad \frac{f : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}) \in \Gamma_1}{\Gamma_1 \vdash f : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (VAR)}}{\Gamma_1 \vdash \langle g, f \rangle : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (PAIR)}$$

Let

$$\Gamma_2 = \Gamma_1, k : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}), \\ p : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))$$

Use $\mathcal{D}_{\text{body}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{first}} \quad \mathcal{D}_{\text{second}}}{\Gamma_2 \vdash \langle k(\text{proj1 } p), \text{proj2 } p \rangle : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (PAIR)}$$

Use $\mathcal{D}_{\text{first}}$ to denote the following derivation tree:

$$\frac{\frac{k : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}) \in \Gamma_2}{\Gamma_2 \vdash k : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (VAR)} \quad \frac{\frac{p : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})) \in \Gamma_2}{\Gamma_2 \vdash p : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (VAR)} \quad \frac{\Gamma_2 \vdash \text{proj1 } p : \mathbf{T} \rightarrow \mathbf{T}}{\Gamma_2 \vdash k(\text{proj1 } p) : \mathbf{T} \rightarrow \mathbf{T}} \text{ (APP)}}{\Gamma_2 \vdash k(\text{proj1 } p) : \mathbf{T} \rightarrow \mathbf{T}} \text{ (PROJ1)}$$

Use $\mathcal{D}_{\text{second}}$ to denote the following derivation tree:

$$\frac{\frac{p : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})) \in \Gamma_2}{\Gamma_2 \vdash p : (\mathbf{T} \rightarrow \mathbf{T}) \times ((\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))} \text{ (VAR)}}{\Gamma_2 \vdash \text{proj2 } p : (\mathbf{T} \rightarrow \mathbf{T}) \rightarrow (\mathbf{T} \rightarrow \mathbf{T})} \text{ (PROJ2)}$$

(b)(ii) Reduction

$$\begin{aligned} U &\longrightarrow \text{let } \langle k, p \rangle = \langle \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h, \langle \lambda x : \mathbf{T}. x, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle \rangle \\ &\quad \text{in } \langle k(\text{proj1 } p), \text{proj2 } p \rangle \\ &\longrightarrow \langle (\lambda h : \mathbf{T} \rightarrow \mathbf{T}. h)(\text{proj1 } \langle \lambda x : \mathbf{T}. x, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle), \\ &\quad \text{proj2 } \langle \lambda x : \mathbf{T}. x, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle \rangle \\ &\longrightarrow \langle (\lambda h : \mathbf{T} \rightarrow \mathbf{T}. h)(\lambda x : \mathbf{T}. x), \\ &\quad \text{proj2 } \langle \lambda x : \mathbf{T}. x, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle \rangle \\ &\longrightarrow \langle \lambda x : \mathbf{T}. x, \text{proj2 } \langle \lambda x : \mathbf{T}. x, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle \rangle \\ &\longrightarrow \langle \lambda x : \mathbf{T}. x, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle \end{aligned}$$

Problem 2 (continued)

(c) Preservation and progress

Variant	Preservation	Progress
1	Holds	Holds
2	Holds	Holds
3	Holds	Fails
4	Fails	Holds
5	Fails	Holds

Variant 3. Let M_3 be the following term:

$$\begin{array}{l} \text{let } \langle f, g \rangle = \langle (\lambda h : \mathbf{T} \rightarrow \mathbf{T}. h)(\lambda z : \mathbf{T}. z), \lambda z : \mathbf{T}. z \rangle \\ \text{in } f \end{array}$$

M_3 is well-typed:

$$\bullet \vdash M_3 : \mathbf{T} \rightarrow \mathbf{T}$$

M_3 is not a value. Its first pair component is not a value, and removing LET-M leaves no applicable reduction rule.

Variant 4. Let M_4 be the following term:

$$\text{let } \langle f, g \rangle = \langle \lambda z : \mathbf{T}. z, \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \rangle \text{ in } f$$

M_4 is well-typed:

$$\bullet \vdash M_4 : \mathbf{T} \rightarrow \mathbf{T}$$

However,

$$M_4 \longrightarrow \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h \quad \bullet \not\vdash \lambda h : \mathbf{T} \rightarrow \mathbf{T}. h : \mathbf{T} \rightarrow \mathbf{T}$$

Variant 5. Let M_5 be the following term:

$$\lambda y : \mathbf{T}. \text{let } \langle x, y \rangle = \langle \lambda z : \mathbf{T}. y, \lambda w : \mathbf{T}. w \rangle \text{ in } x$$

M_5 is well-typed:

$$\bullet \vdash M_5 : \mathbf{T} \rightarrow (\mathbf{T} \rightarrow \mathbf{T})$$

However,

$$\begin{aligned} M_5 &\longrightarrow \lambda y : \mathbf{T}. (x[(\lambda z : \mathbf{T}. y)/x][(\lambda w : \mathbf{T}. w)/y]) \\ &= \lambda y : \mathbf{T}. ((\lambda z : \mathbf{T}. y)[(\lambda w : \mathbf{T}. w)/y]) \\ &= \lambda y : \mathbf{T}. \lambda z : \mathbf{T}. \lambda w : \mathbf{T}. w \end{aligned}$$

The reduct has type $\mathbf{T} \rightarrow (\mathbf{T} \rightarrow (\mathbf{T} \rightarrow \mathbf{T}))$, not $\mathbf{T} \rightarrow (\mathbf{T} \rightarrow \mathbf{T})$.